# Algebraic Manipulation (Simplifying Expressions)

1. June 15/4H/Q7 a & d

| | | | | | |
|---|---|---|---|---|---|
| 7 (a) | | $24 - 18y$ | 1 | B1 | |
| (d) | $y^2 + 10y - 2y - 20$ | | 2 | M1 | for 3 correct terms out of 4<br>or for 4 correct terms ignoring signs<br>or for $y^2 + 8y + c$ for any non-zero value of $c$<br>or for ... $+ 8y - 20$ |
| | | $y^2 + 8y - 20$ | | A1 | cao |

2. June 15/4HR/Q9 a

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|---|---|---|---|---|
| | $10p - 15$ | 1 | B1 | Accept $10 \times p - 15$ |

3. Jan 16/3H/Q4 c

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|---|---|---|---|---|
| $x^2 + 5x - x - 5$ | | | M1 | for 3 correct terms out of a maximum of 4 terms<br>or for 4 correct terms ignoring signs<br>or for $x^2 + 4x + k$ for any non-zero value of $k$<br>or for … $+ 4x - 5$ |
| | $x^2 + 4x - 5$ | 2 | A1 | |

4. Jan 16/3HR/Q2 b

| | | | | |
|---|---|---|---|---|
| $t^2 - 4t + 5t - 20$ | | | M1 | For three correct terms out of four or for four terms correct except for signs. |
| | $t^2 + t - 20$ | 2 | A1 | |

5. Jan 16/4H/Q9 a

| | | | | |
|---|---|---|---|---|
| | $x^2 + 2x$ | 1 | B1 | $2x + x^2$ or $x2 + x^2$ |

6. Jan 16/4HR/Q12

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|---|---|---|---|---|---|
| $(4x^2+6x+6x+9)-(4x^2-6x-6x+9)$ | | | M1 | any one correct expansion | M1 for $((2x+3)+(2x-3))((2x+3)-(2x-3))$ |
| $4x^2+6x+6x+9-4x^2+6x+6x-9$ | | | M1 | for dealing correctly with the negative sign | M1 for $(2x+3+2x-3)(2x+3-2x+3)$ |
| | $24x$ | 3 | A1 | | |

7. June 16/3H/Q7 b

| | | | | |
|---|---|---|---|---|
| | | | M1 | for 3 correct terms **or** 4 correct terms ignoring signs **or** $x^2-7x+a$ for any non-zero value of $a$ **or** $\ldots-7x-18$ |
| | $x^2-7x-18$ | 2 | A1 | |

8. June 16/3HR/Q12 c

| | | | | |
|---|---|---|---|---|
| $8y^2-2y+12y-3$ | $8y^2+10y-3$ | 2 | M1 | For any three correct terms or for 4 correct terms ignoring signs or for $8y^2+10y+k$ for any non-zero value of $k$ or for $\ldots.+10y-3$ |
| | | | A1 | |

9. June 16/4H/Q5 d

| | | | | |
|---|---|---|---|---|
| $8x + 12 + 2x + 10$ | $10x + 22$ | 2 | M1<br>A1 | Any three terms correct out of four.<br>Allow $2(5x + 11)$<br>Do not ISW |

10. June 16/4HR/Q2 b

| | | | | |
|---|---|---|---|---|
| $6y + 6 + 2y - 8$ | $8y - 2$ | 2 | M1<br>A1 | For 3 correct terms<br>oe eg 2(4y – 1) |

11. June 16/4HR/ Q14 c

| | | | | |
|---|---|---|---|---|
| $4x^2 + 12x$ or $4x^2 - 12x + 9$ or $-4x^2 + 12x - 9$ oe | $24x - 9$ or $3(8x - 3)$ | 3 | M1 | For expansion of $4x(x + 3)$ or $(2x - 3)^2$ or $-(2x - 3)^2$ |
| $4x^2 + 12x - 4x^2 + 12x - 9$ | | | M1 | Fully correct expansions with correct removal of bracket (ie all signs correct) |
| | | | A1 | |

12. Jan 17/3H/Q5 b

| | | | | |
|---|---|---|---|---|
| $4p + 20 + 7p - 14$ | | 2 | M1 | Any 3 terms correct |
| | $11p + 6$ | | A1 | cao<br>NB $11p + 6$ followed by, for example, $17p$ scores M1 A0 |

13. Jan 17/3H/Q9

| | | | | |
|---|---|---|---|---|
| $4x^2 + 6x + 6x + 9$ **or** $4x^2 + 12x + 9$ | | 3 | M1 | for at least 3 terms correct in expansion of first pair of brackets |
| $2x^2 - 10x + 3x - 15$ **or** $2x^2 - 7x - 15$ | | | M1 | for at least 3 terms correct in expansion of second pair of brackets **or** all 4 terms correct ignoring signs<br><br>allow $-2x^2 - 7x - 15$ |
| | $2x^2 + 19x + 24$ | | A1 | |
| **Alternative method** | | | | |
| $(2x + 3)[(2x + 3) - (x - 5)]$ | | | M1 | |
| $(2x + 3)(x + 8)$ | | | M1 | |
| | $2x^2 + 19x + 24$ | | A1 | |

14. Jan 17/3HR/Q11 b

| | | | | |
|---|---|---|---|---|
| $k^2 + 9k - 5k - 45$ | | 2 | M1 | for 3 terms correct **or** all 4 terms correct ignoring signs **or** $y^2 + 4k + \ldots$ **or**… $+ 4k - 45$ |
| | $k^2 + 4k - 45$ | | A1 | |

15. Jan 17/4HR/Q7 a

| | | | |
|---|---|---|---|
| | $12p + 15$ | 1 | B1 |

**16. Jan 18/4H/Q10 b & d**

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|---|---|---|---|---|---|
| (b) | | $2p^2 - 3p$ | 1 | B1 | |
| (d) | $x^2 + 7x - 3x - 21$ | $x^2 + 4x - 21$ | 2 | M1 | 3 terms correct with signs **or** 4 terms correct ignoring signs **or** $x^2 + 4x \pm \ldots$. **or** $\ldots\ldots + 4x - 21$ |
| | | | | A1 | |

**17. Jan 18/4HR/Q7 c**

| | | | | |
|---|---|---|---|---|
| $x^2 + 3x - 6x - 18$ | $x^2 - 3x - 18$ | 2 | M1 | For 3 correct terms or<br>For 4 correct terms ignoring signs or<br>For $x^2 - 3x + c$ for any non-zero value of $c$ or<br>For ... $- 3x - 18$ |
| | | | A1 | |

18. June 18/1H/Q11 a

| | | | | |
|---|---|---|---|---|
| $2x^2 - x + 6x - 3$ **or** $2x^2 + 5x - 3$ **or** $x^2 + 3x - 5x - 15$ **or** $x^2 - 2x - 15$ **or** $2x^2 - 10x - x + 5$ **or** $2x^2 - 11x + 5$<br><br>eg.<br>$2x^3 + 5x^2 - 3x - 10x^2 - 25x + 15$ **or**<br>$2x^3 - 4x^2 - 30x - x^2 + 2x + 15$ **or**<br>$2x^3 - 11x^2 + 5x + 6x^2 - 33x + 15$ | $2x^3 - 5x^2 - 28x + 15$ | 3 | M1<br><br>M1<br><br>A1 | for expansion of any 2 of the 3 brackets (at least 3 of 4 terms correct)<br><br>(dep) ft for at least half of their terms correct in second expansion (the correct number of terms **must** be present) |
| **Alternative scheme** | | | | |
| $2x^3 - 10x^2 - x^2 + 5x + 6x^2 - 30x - 3x + 15$ | $2x^3 - 5x^2 - 28x + 15$ | 3 | M2<br><br>A1 | for a complete expansion with 8 terms present, at least 4 of which must be correct |

19. June 18/1HR/Q14 b

| | | | | |
|---|---|---|---|---|
| $3x^2 + 9xy - 4yx - 12y^2$ | $3x^2 + 5xy - 12y^2$ | 2 | M1 | M1 for 3 correct terms out of 4<br>**or** for 4 correct terms ignoring signs<br>**or** for $3x^2 + 5xy + c$ for any non zero value of $c$<br>**or** for $d + 5xy - 12y^2$ for any non zero value of $d$ |
| | | | A1 | |

20. Jan 19/1H/Q1 b

| | | | | |
|---|---|---|---|---|
| $e^2+3e-5e-15$ | | | M1 | for 3 correct terms **or**<br>for 4 correct terms ignoring signs<br>**or** $e^2-2e+k$ for non-zero $k$<br>**or** $\ldots-2e-15$ |
| | $e^2-2e-15$ | 2 | A1 | |

21. Jan 19/1HR/Q15 b

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|---|---|---|---|---|
| $4n(n^2+2n-15)$ **or** $(4n^2-12n)(n+5)$ **or** $(4n^2+20n)(n-3)$ | $4n^3+8n^2-60n$ | 2 | M1 | For a correct partial expansion ( may be unsimplified e.g $4n(n^2+5n-3n-15)$ ) |
| | | | A1 | |

22. June 19/1H/Q3 c

| | | | | |
|---|---|---|---|---|
| $x^2+9x-2x-18$ | | 2 | M1 | for 3 correct terms **or**<br>4 correct terms ignoring signs **or**<br>$x^2+7x+c$ **or** .... $+7x-18$ |
| | $x^2+7x-18$ | | A1 | |

23. June 19/1HR/Q18 a

| | | | | |
|---|---|---|---|---|
| $(x + 2)(2x + 3) = 2x^2 + 3x + 4x + 6$<br>$(2x + 3)(x - 7) = 2x^2 - 14x + 3x - 21$<br>$(x + 2)(x - 7) = x^2 - 7x + 2x - 14$ | | 3 | M1 | For multiplying a pair of brackets and getting 3 out of 4 terms correct. |
| $(2x^2+7x+6)(x-7) = 2x^3-14x^2+7x^2-49x+6x-42$<br>$(2x^2-11x-21)(x+2) = 2x^3+4x^2-11x^2-22x-21x-42$<br>$(x^2-5x-14)(2x+3) = 2x^3+3x^2-10x^2-15x-28x-42$ | | | M1dep | For multiplying the product of the first 2 brackets (ft from the 1st stage) by the 3rd bracket, and getting at least 3 out of 6 or 4 out of 8 terms correct |
| | $2x^3 - 7x^2 - 43x - 42$ | | A1 | Fully correct. isw extra work as long as correct<br>e.g. $x(2x^2 - 7x - 43) - 42$ |
| **Alternative (all in one method)** | | | | |
| $(x + 2)(2x + 3)(x - 7) =$<br>$2x^3 - 14x^2 + 3x^2 - 21x + 4x^2 - 28x + 6x - 42$ | | | M2<br>(M1) | For at least 6 out of 8 correct terms<br>for 4 or 5 out of 8 correct terms |
| | $2x^3 - 7x^2 - 43x - 42$ | | A1 | |

24. Jan 20/1H/Q7 a

| | | | |
|---|---|---|---|
| $m^2 - 8m + 5m - 40$ | | 2 | M1 for any 3 correct terms **or** for 4 out of 4 correct terms ignoring signs<br>for $m^2 - 3m \ldots$ **or**<br>for $\ldots - 3m - 40$ |
| | $m^2 - 3m - 40$ | | A1 |

25. Jan 20/1HR/Q3 c

| | | | | |
|---|---|---|---|---|
| $x^2 - 5x + 2x - 10$ | | | M1 | for any 3 correct terms **or** for 4 out of 4 correct terms ignoring signs **or** $x^2 - 3x\ldots$ **or** for $\ldots -3x - 10$ |
| | $x^2 - 3x - 10$ | 2 | A1 | |

26. June 20/1HR/Q14

| | | | | |
|---|---|---|---|---|
| $(4x+1)(x-3) = 4x^2 - 12x + x - 3 (= 4x^2 - 11x - 3)$<br>$(4x+1)(5x+6) = 20x^2 + 24x + 5x + 6 (= 20x^2 + 29x + 6)$<br>$(x-3)(5x+6) = 5x^2 + 6x - 15x - 18 (= 5x^2 - 9x - 18)$ | | 3 | M1 | for multiplying 2 brackets with at least 3 out of 4 terms correct |
| $(5x+6)(4x^2 - 11x - 3) = 20x^3 - 55x^2 - 15x + 24x^2 - 66x - 18$<br>$(x-3)(20x^2 + 29x + 6) = 20x^3 + 29x^2 + 6x - 60x^2 - 87x - 18$<br>$(4x+1)(5x^2 - 9x - 18) = 20x^3 - 36x^2 - 72x + 5x^2 - 9x - 18$ | | | M1 | (dep) for multiplying the product of the first 2 brackets (ft from the 1st stage) by the 3rd bracket, and getting at least 3 out of 6 or 4 out of 8 terms correct |
| | $20x^3 - 31x^2 - 81x - 18$ | | A1 | |
| **Alternative** | | | | |
| $20x^3 + 24x^2 - 60x^2 + 5x^2 - 15x + 6x - 72x - 18$ | | | B2 | for at least 6 out of 8 terms correct<br>(B1 for 4 or 5 out of 8 correct terms) |
| | $20x^3 - 31x^2 - 81x - 18$ | | A1 | |

27. Jan 21/1H/Q12 b

| | | | | |
|---|---|---|---|---|
| $2x(x-5)=2x^2-10x$<br>**or** $2x(x-3)=2x^2-6x$<br>**or** $(x-5)(x-3)=x^2-5x-3x+15\left(=x^2-8x+15\right)$ | | 3 | M1 | for multiplying $2x$ by a bracket with both terms correct **or** the 2 brackets with at least 3 out of 4 terms correct or at least 2 out of 3 terms correct |
| $(2x^2-10x)(x-3)=2x^3-6x^2-10x^2+30x$<br>**or** $(2x^2-6x)(x-5)=2x^3-10x^2-6x^2+30x$<br>**or** $2x(x^2-5x-3x+15)=2x^3-10x^2-6x^2+30x$<br>**or** $2x(x^2-8x+15)=2x^3-16x^2+30x$ | | | M1 | (dep) for multiplying the product of $2x$ and the 1st bracket (ft from the 1st stage) by the 2nd bracket and getting at least 3 out of 4 terms correct<br>**or** multiplying the product of the 2 brackets (ft from the 1st stage) by the $2x$, and getting at least 3 out of 4 or 2 out of 3 terms correct |
| | $2x^3-16x^2+30x$ | | A1 | |

28. Jan 21/1HR/Q16 a

| | | | | |
|---|---|---|---|---|
| E.g.<br>$x^2 + 4x - 2x - 8$ $(= x^2 + 2x - 8)$<br><br>**or**<br><br>$x^2 - 2x + x - 2$ $(= x^2 - x - 2)$<br><br>**or**<br><br>$x^2 + 4x + x + 4$ $(= x^2 + 5x + 4)$ | | 3 | M1 | for multiplying out two brackets correctly with no more than one error |
| E.g.<br>$x^3 + 2x^2 - 8x + x^2 + 2x - 8$ or<br>$x^3 + 4x^2 - 2x^2 - 8x + x^2 + 4x - 2x - 8$<br><br>**or**<br><br>$x^3 - x^2 - 2x + 4x^2 - 4x - 8$ or<br>$x^3 - 2x^2 + x^2 - 2x + 4x^2 - 8x + 4x - 8$<br><br>**or**<br><br>$x^3 + 5x^2 + 4x - 2x^2 - 10x - 8$ or<br>$x^3 + 4x^2 + x^2 + 4x - 2x^2 - 8x - 2x - 8$ | | | M1 | for at least 3 terms correct out of a maximum of 6 terms<br><br>**or**<br><br>for at least 4 terms correct out of a maximum of 8 terms |
| | $x^3 + 3x^2 - 6x - 8$ | | A1 | |

29. Jan 21/2H/Q5 a

| | | | | |
|---|---|---|---|---|
| $8x^2 + 20x - 6x^2 + 9x$ | $2x^2 + 29x$ | 2 | M1 | 3 correct terms or all 4 terms condoning incorrect signs |
| | | | A1 | |

30. June 21/1H/Q13

| | | | | |
|---|---|---|---|---|
| $4x(3x+1)=12x^2+4x$<br>**or** $4x(2x-3)=8x^2-12x$<br>**or** $(3x+1)(2x-3)=6x^2-9x+2x-3\ (=6x^2-7x-3)$ | | 3 | M1 | for expanding two of the three factors, allow one error |
| $(12x^2+4x)(2x-3)=24x^3-36x^2+8x^2-12x$ oe<br>$(8x^2-12x)(3x+1)=24x^3+8x^2-36x^2-12x$ oe<br>$4x(6x^2-7x-3)=$ eg $24x^3-28x^2....$ oe | | | M1 | (dep)ft for expanding by the third factor, allow one error<br>(some may do the expansion in one stage and will get to $24x^3-36x^2+8x^2-12x$ without firstly expanding two factors) |
| *Working required* | $24x^3-28x^2-12x$ | | A1 | dep on M1<br><br>isw correct factorisation eg<br>$4(6x^3-7x^2-3x)$<br>$x(24x^2-28x-12)$<br>$4x(6x^2-7x-3)$<br><br>do not isw incorrect simplification eg $24x^3-28x^2-12x=6x^3-7x^2-3x$ gets M2A0 |

31. Nov 21/2H/Q12 c

| | | | |
|---|---|---|---|
| eg<br>$4x(x-5) = 4x^2 - 20x$ or<br>$4x(2x+3) = 8x^2 + 12x$ or<br>$(x-5)(2x+3) = 2x^2 + 3x - 10x - 15$<br>$= 2x^2 - 7x - 15$ | | 3 | M1 allow one error in the expansion of<br>$4x(x-5)$ or<br>$4x(2x+3)$ or<br>$(x-5)(2x+3)$ |
| eg<br>$(4x^2 - 20x)(2x+3) = 8x^3 + 12x^2 - 40x^2 - 60x$ or<br>$(8x^2 + 12x)(x-5) = 8x^3 + 12x^2 - 40x^2 - 60x$<br>or<br>$4x(2x^2 + 3x - 10x - 15) = 8x^3 + 12x^2 - 40x^2 - 60x$<br>or<br>$4x(2x^2 - 7x - 15) = 8x^3 - 28x^2 - 60x$ | | | M1ft but dep on previous M1 for correctly expanding – allow one extra error or one omission. |
| *Working required* | $8x^3 - 28x^2 - 60x$ | | A1 dep on M1<br>May be factorised if $8x^3 - 28x^2 - 60x$ seen |

32. Jan 22/1H/Q13 a

| | | | | |
|---|---|---|---|---|
| $5x(x+2)=5x^2+10x$<br>**or** $(x+2)(3x-4)=3x^2-4x+6x-8(=3x^2+2x-8)$<br>**or** $5x(3x-4)=15x^2-20x$ | | 3 | M1 | for a correct intention to multiply all 3 factors by starting to multiply 2 factors only, allow one error |
| eg<br>$[(5x^2+10x)(3x-4)=]\ 15x^3-20x^2+30x^2-40x$<br>**or** $[5x(3x^2+2x-8)=]\ 15x^3+10x^2-40x$<br>**or** $[(x+2)(15x^2-20x)=]\ 15x^3-20x^2+30x^2-40x$ | | | M1 | (dep)ft for expanding by the third factor, allow one error<br><br>(some may do the expansion in one stage and will get to $15x^3-20x^2+30x^2-40x$ without firstly expanding two factors, allow two errors) |
| | $15x^3+10x^2-40x$ | | A1 | isw correct factorisation<br>eg $5(3x^3+2x^2-8x)$<br><br>do not isw incorrect factorisation<br>eg $15x^3+10x^2-40x = 3x^3+2x^2-8x$ |

33. Jan 22/1HR/Q12 a

| | | | |
|---|---|---|---|
| $n(3n^2+5n-12n-20)$ **or** $n(3n^2-7n-20)$ **or** $(3n^2+5n)(n-4)$ **or** $(n^2-4n)(3n+5)$ **or** $3n^3+5n^2-12n^2-20n$ | | 2 | M1 for a correct partial expansion (may be unsimplified) (allow one error in the expansion of $(n-4)(3n+5)$ e.g.<br>for any 3 correct terms<br>**or**<br>for 4 out of 4 correct terms ignoring signs **or** for $3n^2-7n\ldots$<br>**or**<br>for $\ldots-7n-20)$ |
| | $3n^3-7n^2-20n$ | | A1 oe e.g. if correct answer seen allow further factorisation to $n(3n^2-7n-20)$ |

34. Jan 23/1H/Q13

| | | | |
|---|---|---|---|
| $3x(2x-5)=6x^2-15x$ **or** | | 3 | M1 for multiplying $3x$ by $(2x-5)$ with **both** terms correct or |
| $(2x-5)^2=4x^2-10x-10x+25$ **or** | | | for multiplying $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or |
| $(2x-5)^2=4x^2-20x+25$ | | | for multiplying $(2x-5)$ by $(2x-5)$ and getting $4x^2-20x\ldots$ **or** $\ldots-20x+25$ (not for $4x^2+25$) |
| $(6x^2-15x)(2x-5)=12x^3-30x^2-30x^2+75x$ oe **or** | | | M1ft (dep) for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or |
| $(6x^2-15x)(2x-5)=12x^3-60x^2+75x$ oe **or** | | | for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ and getting $12x^3-60x^2\ldots$ **or** $\ldots-60x^2+75x$ |
| $3x(4x^2-10x-10x+25)=12x^3-30x^2-30x^2+75x$ oe **or** | | | for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 3 out of 4 terms correct or |
| $3x(4x^2-20x+25)=12x^3-60x^2+75x$ | | | for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 2 out of 3 terms correct or |
| | | | Expansion in one stage will lead to $12x^3-30x^2-30x^2+75x$ without firstly expanding two factors – award M2 for 3 out of 4 terms correct M1 for 2 out of 4 terms correct |
| *Working required* | $12x^3-60x^2+75x$ | | A1 dep on M1 |

35. Jan 23/2HR/Q12 c

| | | | | |
|---|---|---|---|---|
| $5x(3x+4)=15x^2+20x$<br>**or** $5x(2x-1)=10x^2-5x$<br>**or** $(3x+4)(2x-1)=6x^2-3x+8x-4$<br>$(=6x^2+5x-4)$ | | 3 | M1 | for a correct intention to multiply all 3 factors by multiplying 2 factors only, allow one error |
| $(15x^2+20x)(2x-1)=30x^3-15x^2+40x^2-20x$ oe<br>$(10x^2-5x)(3x+4)=30x^3+40x^2-15x^2-20x$ oe<br>$5x(6x^2+5x-4)=30x^3+25x^2-20x$ oe | | | M1 | (dep)ft for expanding by the third factor, allow one error<br>(some may do the expansion in one stage and will get to $30x^3-15x^2+40x^2-20x$ without firstly expanding two factors – this gains M2, allow one error) |
| *Correct answer scores full marks (unless from obvious incorrect working)* | $30x^3+25x^2-20x$ | | A1 | isw correct factorisation ($30x^3+25x^2-20x$ must be seen previously to award 3 marks)<br>eg<br>$5(6x^3+5x^2-4x)$<br>$x(30x^2+25x-20)$<br>$5x(6x^2+5x-4)$<br><br>do not isw incorrect simplification<br>eg $30x^3+25x^2-20x=6x^3+5x^2-4x$<br>gets M2A0 |

36. June 23/2H/Q17 a

| | | | | |
|---|---|---|---|---|
| eg $3x^2-2x+18x-12$ $(=3x^2+16x-12)$<br>**or** $x^2+6x+6x+36$ $(=x^2+12x+36)$<br>(allow in a table with no sign indicating +) | | 3 | M1 | for a correct method to expand two brackets with at least 3 terms correct out of 4 terms seen (or 2 terms correct out of 3 terms seen) |
| eg $3x^3+36x^2+108x-2x^2-24x-72$ or<br>$3x^3+18x^2+18x^2+108x-2x^2-12x-12x-72$<br>**or**<br>$3x^3+16x^2-12x+18x^2+96x-72$ or<br>$3x^3-2x^2+18x^2-12x+18x^2-12x+108x-72$ | | | M1ft | ft dep on M1 and a quadratic for a correct method to multiply by the 3rd bracket allow one further error |
| *Correct answer scores full marks (unless from obvious incorrect working)* | $3x^3+34x^2+84x-72$ | | A1 | If no working shown then award B2 for 3 out of a maximum of 4 terms correct |
| ***ALTERNATIVE*** | | | | |
| $3x^3-2x^2+18x^2-12x+18x^2-12x+108x-72$ | | 3 | M2 | For a complete expansion with 8 terms present of which 4 are correct<br>(M1 for 4 correct terms from any number of terms) |
| | $3x^3+34x^2+84x-72$ | | A1 | |
| See next page for 17(b) | | | | |

## 37. June 24/1HR/Q 14

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|---|---|---|---|---|---|
| **14** (a) | $(3x+1)(2-x) = 6x - 3x^2 + 2 - x\ (= -3x^2 + 5x + 2)$<br>**or**<br>$(2-x)(4+x) = 8 + 2x - 4x - x^2\ (= -x^2 - 2x + 8)$<br>**or**<br>$(3x+1)(4+x) = 12x + 3x^2 + 4 + x\ (= 3x^2 + 13x + 4)$ | | 3 | M1 | for a correct method to expand two brackets with at least 3 terms correct out of 4 terms (or 2 terms correct out of 3 terms). Do not award this mark for eg $6x - 3x^2 + 2 - x + 8 + 2x - 4x - x^2$ or $6x - 3x^2 + 2 - x + 4 + x$ |
| | $(-3x^2 + 5x + 2)(4 + x) = -12x^2 - 3x^3 + 20x + 5x^2 + 8 + 2x$<br>**or**<br>$(-x^2 - 2x + 8)(3x + 1) = -3x^3 - x^2 - 6x^2 - 2x + 24x + 8$<br>**or**<br>$(3x^2 + 13x + 4)(2 - x) = 6x^2 - 3x^3 + 26x - 13x^2 + 8 - 4x$ | | | M1 | ft dep on M1 **and** a quadratic for a correct method to multiply by the 3rd bracket – allow one further error |
| | *Correct answer scores full marks (unless from obvious incorrect working)* | $-3x^3 - 7x^2 + 22x + 8$ | | A1 | oe but must be simplified<br>eg $22x - 3x^3 - 7x^2 + 8$<br>if no working shown then award B2 for 3 out of a maximum of 4 terms correct |
| (b) | $\left(\frac{1}{a^6b^4}\right)^{-\frac{1}{2}}$ **or** $\left(a^{-6}b^{-4}\right)^{-\frac{1}{2}}$ **or** $\left(\frac{a^{1.5}b^{0.5}}{a^{4.5}b^{2.5}}\right)^{-1}$<br>**or** $\left(\frac{a^9b^5}{a^3b}\right)^{\frac{1}{2}}$ **or** $\left(\frac{a^{-3}b^{-1}}{a^{-9}b^{-5}}\right)^{\frac{1}{2}}$ oe | | 3 | M1 | for simplifying the $a$ and the $b$ term in the fraction<br>**or** for applying the power ½ to at least 3 out of 4 of $a^3$, $b$, $a^9$, $b^5$<br>**or** for applying the negative power to at least 3 out of 4 of $a^3$, $b$, $a^9$, $b^5$ |
| | $\left(\frac{1}{a^3b^2}\right)^{-1}$ **or** $\left(a^{-3}b^{-2}\right)^{-1}$ **or** $\frac{a^{4.5}b^{2.5}}{a^{1.5}b^{0.5}}$<br>**or** $\left(a^6b^4\right)^{\frac{1}{2}}$ **or** $\left(\frac{a^{-1.5}b^{-0.5}}{a^{-4.5}b^{-2.5}}\right)$ oe | | | M1 | for two of<br>simplifying the $a$ and the $b$ term in the fraction<br>**or** for applying the power ½ to at least 3 out of 4 of $a^3$, $b$, $a^9$, $b^5$<br>**or** for applying the negative power to at least 3 out of 4 of $a^3$, $b$, $a^9$, $b^5$ |
| | *Correct answer scores full marks (unless from obvious incorrect working)* | $a^3b^2$ | | A1 | accept $\frac{1}{a^{-3}b^{-2}}$ |
| | | | | | **Total 6 marks** |